Token-Efficient Context Sharing in Multi-Agent LLM Systems: How Much Can Be Cut, and What Limits It

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Abstract

Multi-agent LLM systems coordinate by forwarding context between agents, and every forwarded token is billed. Common practice forwards whole conversation histories, because deciding what to withhold is hard, so coordination cost grows quadratically in the number of agents. We ask a practical question: how much of that traffic can be removed without paying for it in task quality, and what stops us removing more?

On a controlled proxy for multi-agent context sharing built from a public multi-hop benchmark, a 2.7× reduction in inter-agent tokens has an observed cost of 4.2 accuracy points, indistinguishable from zero at this sample size (95% CI crossing zero); an oracle reaches 21× fewer tokens with an observed +5-point *gain*, significant on one of two models tested. Over-delivery may thus be mildly harmful rather than merely wasteful, with roughly an order of magnitude available in principle, though the evidence for that ceiling is directional. A practical policy at 5.4× does cost 8 points with a CI excluding zero, so the frontier is real.

How the budget is spent matters more than how much is spent. Allocating per receiver beats selecting one globally-scored set for everyone by 19.2 accuracy points at matched cost, and we explain this with a separation result: the one-set formulation — the natural reading of the problem as influence maximization — loses a factor n in the number of receivers, by two independent constructions, even on instances where the objective is monotone and submodular; adding saturation removes any finite bound. We also show the two nearest optimisation toolchains do not apply here, the second because its structural parameters do not exist rather than because its bound is loose — and, on the positive side, that the correct per-receiver formulation costs nothing extra in approximability under an explicit oracle assumption, since it decomposes exactly into a resource-allocation program over a shared budget. Abandoning the one-set object is therefore not a tractability trade.

Finally, a negative result that we think is the most useful finding for practitioners. Across four allocation algorithms, including a resource-allocation program that exploits structure no baseline can, none beats a tuned uniform per-receiver rule at matched budget; and replacing lexical relevance with a dense retriever makes matters worse. Three independent lines of evidence place the binding constraint on *estimating* which agent needs what, not on optimising the allocation. The remaining order of magnitude is a retrieval problem, and effort spent on cleverer allocation will not recover it.

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Keywords

multi-agent systems; large language models; communication; resource allocation; submodular optimization

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1 Introduction

A multi-agent LLM system coordinates by passing context. A planner tells a solver what the constraints are; an auditor is handed the records it must check. Every such transmission is billed per token, so coordination is a direct operating cost, and the prevailing engineering answer — forward the whole conversation, because deciding what to withhold is hard and sending everything is safe — makes that cost grow quadratically in the number of agents for architectures that accumulate or broadcast history (sparse topologies or prefix caching scale differently). We use transmitted tokens as a proxy for operating cost throughout; real billing also depends on input/output pricing, caching, and model choice.

This paper treats that as the problem to solve. We ask how much inter-agent traffic can be removed before task quality suffers, what the shape of the cost–quality frontier is, and what prevents us from reaching the cheap allocations that clearly exist.

What we find. Three things, in decreasing order of how comfortable they are.

First, **a lot of the traffic looks removable at no measured cost**. On a controlled proxy built from a public multi-hop benchmark, cutting inter-agent tokens by 2.7× has an observed cost of 4.2 accuracy points that our sample cannot distinguish from zero, and an oracle allocation — which knows which item each agent needs — reaches 21× fewer tokens with an observed +5-point *gain*, significant on one of the two models we test. Over-delivery is therefore plausibly mildly harmful and not merely expensive, which would be consistent with the measured tendency of LLMs to degrade under irrelevant context [18, 19, 25], though we did not run a pre-registered non-inferiority test and say so explicitly (§6.1). An order of magnitude is available in principle.

Second, **how the budget is spent matters more than how much**. Allocating a possibly different set to each receiver beats choosing one globally-scored set for everyone by 19.2 accuracy points at matched cost. This is not a tuning detail: we prove the one-set formulation loses a factor n in the number of receivers (Proposition 1), by two independent constructions, on instances where the objective is monotone and submodular; once saturation is added, the same family of instances admits no finite bound at all. That formulation is the natural reading of context sharing as influence maximization [15], so a substantial and otherwise applicable literature [1, 7, 23] turns out to optimise the wrong object here. We also show that the nearest general framework for the messy objective this problem actually has is unavailable, because its structural parameters fail to exist rather than merely giving a loose bound (§4.2). The switch of object costs nothing in return: the per-receiver problem decomposes exactly into a resource-allocation program coupled only by the shared budget, and any approximation available for one receiver transfers unchanged to n of them (Corollary 5).

Third, and most useful to anyone building such a system, **the remaining order of magnitude is not an optimisation problem**. We implement four allocation algorithms, including one that exploits separability structure no per-receiver baseline can imitate, and none beats a tuned uniform per-receiver rule at matched budget. Replacing lexical relevance with a dense retriever makes per-receiver delivery *worse*. Three independent observations converge: the binding constraint is estimating which agent needs what. In the composite-receiver design built to expose it (§5), the oracle reaches full delivery recall on 208 tokens where the best implementable policy reaches only 62% recall on 929, and no allocation algorithm closes that gap, because the information required is absent from the scores rather than badly exploited.

Contributions. (i) A measured cost–quality frontier for inter-agent context sharing on a controlled proxy task, including a token reduction indistinguishable from zero accuracy cost at this sample size and the oracle ceiling (§5). (ii) A separation result showing the one-set formulation loses a factor n even on monotone submodular instances and is unbounded once saturation is added (§4.1), together with two degeneracy results removing the standard toolchains (§4.2). (iii) An exact decomposition of the per-receiver problem into a resource-allocation program coupled only by the shared budget, with the consequence that its approximability equals the single-receiver one, under an integer-cost, oracle-access assumption we state precisely (§4.3). (iv) A negative result locating the residual gap in relevance estimation rather than allocation (§5). All theory statements were additionally checked by exhaustive enumeration on small instances.

2 Related work

Influence maximization. Monotonicity and submodularity of the spread function are what license greedy's $(1 - 1/e)$ [15], and the scaling literature [1, 7, 23] inherits those assumptions. Recent work relaxes them in various directions but keeps the seed-set object: one chosen set, serving the whole network. Our separation is orthogonal to the quality of those relaxations.

Non-monotone and non-submodular maximization. Guarantees exist for non-monotone submodular maximization [2, 10], for weakly submodular objectives via the submodularity ratio [5, 6] or supermodular degree [9], and for differences of submodular functions [13]. Closest to our setting, Shi and Lai [22] treat non-monotone, non-submodular maximization under a knapsack. §4.2 shows their parameters do not exist here. Regularized objectives of the form $g - c$ with g submodular and c modular admit distorted-greedy guarantees [12]; the density rule we discuss in §4.3 is an instance of that lineage, and we credit it as such rather than presenting it as new.

Context and communication cost in LLM agents. Empirical work cuts inter-agent tokens by pruning the message graph [26], generating the topology per task [8, 14], or sharing key–value caches instead of text [21]; none states an optimality guarantee, and a recent survey of token cost in agent systems reports no formal treatment of the allocation question either [3]. Closest in spirit to our separation, [20] studies empirically how correct and erroneous outputs propagate under topologies of differing sparsity, finding intermediate sparsity best — consistent with our claim that broadcasting is not the safe default, though reached without an optimization formulation. Closest to our applied setting, RCR-Router selects a memory subset per agent under a strict token budget and reports up to 30% savings on multi-hop QA, again without guarantees [17]. A parallel line does prove guarantees, but for a *single* receiver: PACMS treats context assembly as submodular selection [11], and BPS maximizes a monotone submodular benefit minus a modular token penalty under a hard budget, with a bicriteria $(1 - 1/e, 1)$ ratio [4]. BPS is the single-receiver, modular-penalty, submodular-benefit special case of the problem we study, and the closest prior work; it observes that redundant context "can even degrade performance" and then charges that degradation as a separate, explicitly *first-order* linear cost, keeping the benefit a concave, dimension-additive coverage function — so a pair separately inert but jointly decisive has no representation in either half of its model, which is the case §4.2 turns on. On the theory side, [16] treats finite context windows as hard fan-in limits and locates saturation via a mixing depth, but through majority-vote aggregation rather than set-function optimization. We are not aware of a formal treatment of the multi-receiver token-allocation problem, nor of an influence-maximization framing of LLM-agent communication.

3 Model

Receivers (agents) are indexed $1, \dots, n$. A pool F of information items is given, item f costing $c(f) > 0$ tokens. An *allocation* assigns a set $S_i \subseteq F$ to each receiver. Information is copyable, but each transmission is charged, so total spend is $\sum_i c(S_i)$ and serving n receivers with one set costs n times serving one. We also consider a *free-broadcast* convention, where a set may be given to everyone for one transmission, to show our results do not rest on the cost model.

Definition 1 (Multi-receiver context allocation). *An instance is a tuple $(n, F, c, (u_i)_{i \le n}, B)$ with n receivers, a finite item pool F, a cost $c : F \to \mathbb{R}_{>0}$ extended modularly to sets, per-receiver utilities $u_i : 2^F \to \mathbb{R}$ with $u_i(\emptyset) = 0$, and a budget $B > 0$. A feasible* allocation *is a tuple $(S_1, \dots, S_n) \in (2^F)^n$ with $\sum_i c(S_i) \le B$, and the objective is $U(S_1, \dots, S_n) = \sum_i u_i(S_i)$. The* seed-set restriction *of the same instance requires $S_1 = \cdots = S_n$.*

The seed-set restriction is our analogue, in a setting with no diffusion process, of the commitment an influence-maximization formulation makes: one chosen set as the entire decision variable. Classical IM's diffusion can still *reach* different nodes differently once the seed set is fixed; here, where content is transmitted directly rather than propagated, the corresponding restriction is on the delivered content itself, which is why we state it as a restriction on Definition 1 rather than import an IM algorithm unmodified. Fixing everything else in Definition 1 and varying only this restriction is what makes the comparison in §4.1 a controlled one rather than a change of subject; §5's seed-set baseline is our own translation of the idea (global top-k by aggregate score) rather than a published IM algorithm.

Per-receiver utility is

$$u_i(S) = \mathrm{rel}_i(S) - \rho_i(\mathrm{conf}_i(S))\,, \tag{1}$$

relevance minus a saturation penalty, with $\mathrm{rel}_i(\emptyset) = \mathrm{conf}_i(\emptyset) = \rho_i(0) = 0$ so that $u_i(\emptyset) = 0$ as Definition 1 requires. Two further features of (1) are empirically motivated rather than assumed for convenience. First, ρ_i is increasing: delivering more context degrades an agent [19, 25]. Second, the penalty's argument is *confusable* load conf_i, not raw token count, because length-matched studies find that semantic proximity to the task predicts degradation while raw length barely does [18]. Cost and damage are therefore different quantities in general: one is billed and modular, the other is semantic and per-receiver; §5 states explicitly where the experiments collapse this distinction for lack of a confusability oracle. The objective is $U = \sum_i u_i(S_i)$ subject to the budget.

Example 1. Three agents share a pool of 20 documents. Agent 1 must check a delivery date, agent 2 a payment total, agent 3 must reconcile the two. Agents 1 and 2 each need one document; agent 3 needs both. A uniform allowance of one document per agent starves agent 3; an allowance of two overfeeds agents 1 and 2 with a document that can only distract them. No single set serves all three, and the cheapest correct allocation sends different sets to different agents. This is the structure Proposition 1 makes precise.

Why routing intuitions do not transfer. The problem is priced like a delivery problem, which invites a vehicle-routing analogy that fails on one property: goods are conserved, so a delivered unit cannot serve a second customer, whereas context is *copyable* and one item can serve every receiver at once. Conservation is what makes routing a partition problem; without it, the allocation is a covering problem, and the interesting constraint is who should be *spared* an item rather than who gets the last one. Only *capacity* survives the analogy — a context window is rivalrous the way a vehicle is, which is what ρ_i encodes — and capacity here is soft, per-receiver, and semantic, so a per-vehicle-capacity formulation is not a route to the multi-receiver guarantee we lack.

On the shape of ρ_i. We do not assume ρ_i convex: reported degradation curves are S-shaped, flat then a steep fall [18, 19], and magnitudes are strongly model-dependent (one study finds large models almost unaffected by non-confusable filler). Proposition 4 buys the freedom to avoid that fragility — the penalty enters only through a chosen budget level, so ρ_i may be non-convex, discontinuous, or a cliff. Only the density-greedy *heuristic* of §4.3, not the exact dominance test it approximates, needs ρ_i differentiable, and we flag that there as the heuristic's limitation.

4 Theory: separation, degeneracy, and structure

4.1 The seed-set formulation loses a factor n

Proposition 1. *For every $n \geq 1$ there is an instance with n receivers on which, at an identical budget,*

$$\frac{\mathrm{OPT}_{\textit{per-receiver}}}{\mathrm{OPT}_{\textit{seed-set}}} = n,$$

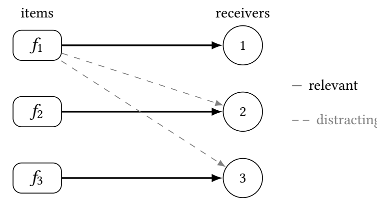

Figure 1: Proposition 1's instance at $n = 3$: each item is relevant to one receiver (solid) and distracting to the other $n - 1$ (dashed, shown only for f_1). Per-receiver follows the solid edges; a seed set must also send down the dashed ones.

by two independent constructions: one from the hard budget alone (cost convention (a) below), one from saturation alone with the budget non-binding (convention (b)). For every $n \geq 2$, the same family of instances under convention (b) admits no finite multiplicative bound at all: the ratio grows without limit as a distraction penalty λ approaches a threshold from below, and at the threshold $\mathrm{OPT}_{\textit{seed-set}}$ is exactly 0 while $\mathrm{OPT}_{\textit{per-receiver}}$ stays at n.

Proof. Take n receivers, n distinct items each of cost c, and let receiver i be satisfied only by item i. Set the budget $B = nc$.

(a) Billed per receiver. Per-receiver allocation sends item i to receiver i, spending $nc = B$ for utility n. A seed set S reaches all n receivers and so costs $n|S|c \leq nc$, forcing $|S| \leq 1$; a single item satisfies exactly one receiver, for utility 1. Ratio n.

(b) Free broadcast, saturating attention. Let the budget be non-binding, let a receiver absorb one item freely, and let each further item cost it 1. Broadcasting S to everyone yields $|S| - n\max(0, |S| - 1)$, maximized at $|S| = 1$ with value 1, while per-receiver allocation again yields n. Ratio n.

(c) Unboundedness, $n \geq 2$. Keep the same items and receivers, keep broadcast free, and let a receiver pay $\lambda \geq 0$ for each item it receives that is irrelevant to it. For a seed set S of size k, every item in S satisfies its own receiver and distracts the other $n - 1$, so summing over all n receivers gives $U_{\mathrm{seed}}(S) = k\big(1 - \lambda(n-1)\big)$, independent of which k items are chosen. For $0 \leq \lambda < 1/(n-1)$ this is increasing in k, so $\mathrm{OPT}_{\text{seed-set}} = n\big(1 - \lambda(n-1)\big) > 0$ and

$$\frac{\mathrm{OPT}_{\text{per-receiver}}}{\mathrm{OPT}_{\text{seed-set}}} = \frac{1}{1 - \lambda(n-1)} \xrightarrow[\lambda \to (1/(n-1))^-]{} \infty,$$

so no λ-independent finite bound covers this family. At $\lambda = 1/(n-1)$ the seed-set value is $U_{\mathrm{seed}}(S) = 0$ for every S, so $\mathrm{OPT}_{\text{seed-set}} = 0$; per-receiver allocation delivers item i to receiver i alone, incurs no penalty, and attains n regardless of λ. □

Case (b) is the more informative one. A reader may discount (a) as an artifact of charging per receiver; (b) removes that objection, because even with free broadcast the seed set is capped. The reason is worth stating plainly: *an additional broadcast item helps one receiver and harms the other $n - 1$.*

Remark 1. The instance in case (a) contains no saturation and no complementarity: rel is modular and $\rho \equiv 0$, so the objective is monotone and submodular. The separation is therefore not a consequence of the degeneracies in §4.2. Repairing the objective's curvature does not rescue the seed-set formulation, because the loss comes from the choice of object, not from the objective's shape. Any guarantee proved against a seed-set optimum bounds distance from the wrong quantity.

Neither construction claims to be the worst case over any class, and we make no such claim: an instance where every receiver needs the same item gives ratio 1 under either convention, so the seed-set loss is not uniformly n across monotone submodular instances, only *at least* n on some of them. What (a)–(c) together show is an existence result on two fronts: the gap is already n on instances as well-behaved as one could ask (modular relevance, no saturation, so U is monotone and submodular by Proposition 3's own terms), and once saturation is strong enough to make broadcasting actively harmful, the same family admits no finite multiplicative bound at all. Both constructions were checked by exhaustive enumeration over small instances rather than by algebra alone.

For completeness we record the hardness of the correct object; it is routine and we claim no credit for it.

Proposition 2. *Maximising U subject to $\sum_i c(S_i) \leq B$ is NP-hard, already for $n = 1$ and $\rho \equiv 0$.*

Proof. With one receiver and no saturation the problem is to choose $S \subseteq F$ with $c(S) \leq B$ maximising a modular rel, which is Knapsack. For general n under one shared budget it contains Multiple Knapsack. □

Remark 2. What Proposition 2 does *not* settle is whether saturation adds hardness beyond the knapsack structure. The construction above switches ρ off, so the hardness is entirely attributable to the budget. We leave the question open and note that Proposition 5 below is evidence in the other direction: the multi-receiver structure adds no approximation hardness at all.

4.2 Both toolchains degenerate

Proposition 3. *Neither hypothesis of the greedy $(1 - 1/e)$ argument, nor either parameter of [22]'s general theorem, holds across the class of Definition 1. Precisely, on two separate witnesses:*

(a) Saturation. *There is an instance with* rel *modular and ρ increasing on which U is not monotone, and no $\gamma_2 \geq 1$ makes U γ_2-weak supermodular in the sense of [22].*

(b) Complementarity. *There is an instance — necessarily with* rel not *modular, since a modular* rel *cannot satisfy the complementarity below — on which U is not submodular, and no finite γ_1 makes U γ_1-weak submodular.*

Because [22]'s Theorem 4 requires γ_1 and γ_2 together, and (a)'s hypothesis of modular rel *is exactly what (b)'s witness must violate, no single instance — let alone a uniform relaxation — can repair both at once.*

Monotonicity and γ_2 fail under saturation. With rel modular and ρ increasing, an item whose relevance falls below its marginal saturation cost strictly lowers u_i, so U is not monotone; this is witness (a).

Submodularity and γ_1 fail under complementarity. Two items individually inert but jointly decisive — $\mathrm{rel}(\emptyset) = \mathrm{rel}(\{a\}) = \mathrm{rel}(\{b\}) = 0$, $\mathrm{rel}(\{a, b\}) = 1$ — give increasing returns, so U is not submodular; this is witness (b). Definition 1 places no modularity requirement on rel_i, so this instance is a legitimate member of the class even though it falls outside witness (a)'s hypothesis. Both failures are required by the greedy argument, on their respective instances.

Table 1: Existence of the two parameters that [22] requires *simultaneously*. Each phenomenon destroys exactly one.

	γ_1 (weak sub.)	γ_2 (weak super.)
Complementarity	**does not exist**	exists
Saturation	exists	**does not exist**

The nearest general framework has no parameters here. Shi and Lai [22] give guarantees for non-monotone, non-submodular maximization under a knapsack. Their general theorem requires the objective to be simultaneously γ_1-weak submodular and γ_2-weak supermodular:

$$F(U_2 \cup \{x\}) - F(U_2) \leq \gamma_1 \left[F(U_1 \cup \{x\}) - F(U_1)\right], \quad (2)$$

$$\gamma_2 \left(F(U_2 \cup \{x\}) - F(U_2)\right) \geq F(U_1 \cup \{x\}) - F(U_1), \quad (3)$$

for $U_1 \subsetneq U_2$ and $x \notin U_2$. Our two phenomena destroy exactly one parameter each.

Under complementarity, take the instance above with $U_1 = \emptyset$, $U_2 = \{b\}$, $x = a$: (2) demands $1 \leq \gamma_1 \cdot 0$, which no finite γ_1 satisfies. Under saturation, an item whose marginal is +0.5 at a lightly loaded receiver and −1.5 at a saturated one makes (3) demand $\gamma_2 \cdot (-1.5) \geq 0.5$, which no $\gamma_2 \geq 1$ satisfies.

Table 1 summarizes, and the witnesses above are its proof, additionally found by exhaustive search over small instances so the failure is not an artifact of a hand-picked example. This is a stronger statement than a loose bound — the parameters are undefined rather than merely weak — but it does not say [22] is wrong: its parameters are exactly right for the class it addresses, and our class simply sits outside that parameterisation, symmetrically on both sides, so no single relaxation recovers the theorem. A framework covering this setting would need unbounded local curvature in both directions at once, which we do not know how to build.

4.3 Structure and algorithms

Proposition 1 says the per-receiver object is the right one. It does not say the resulting problem is intractable, and in one respect it is unusually well structured.

Proposition 4. *The problem is separable across receivers and coupled only by the shared budget. Consequently, writing $v_i(b) = \max\{u_i(S) : c(S) \leq b\}$, an optimal allocation is obtained by solving*

$$\max \left\{ \textstyle\sum_i v_i(b_i) \;:\; \sum_i b_i \leq B \right\}$$

and giving receiver i an optimal set at its level b_i. If each v_i is computed exactly, this is exact.

Proof. $U = \sum_i u_i(S_i)$ and the only constraint linking receivers is $\sum_i c(S_i) \leq B$. Fix any feasible allocation and let $b_i = c(S_i)$. Then $u_i(S_i) \leq v_i(b_i)$ and $\sum_i b_i \leq B$, so its value is at most that of the displayed program. Conversely any feasible (b_i) with maximisers S_i is a feasible allocation of value $\sum_i v_i(b_i)$. □

Two consequences matter. First, the coupling between receivers is a single scalar variable, the shared budget B, and nothing else — no pairwise interaction between receivers survives the reformulation. Second, and less obviously, the saturation penalty enters *only* through the chosen level b_i, so ρ_i may have any shape whatever — non-convex, discontinuous, a cliff. This matters because the measured degradation curves are S-shaped rather than convex, and a rule that reasons about ρ_i' requires convexity to argue that its admission bar rises as a receiver fills.

Remark 3. The scalar coupling is *not*, in general, a market-clearing price: that needs every v_i concave, which Proposition 4 does not assume. Take $v_1(0,1,2) = (0,0,3)$, $v_2(0,1,2) = (0,2,2)$, $B = 2$: the optimum spends all of B on receiver 1, value 3. For $\lambda < 1.5$ the local optima $\arg\max_b v_i(b) - \lambda b$ sum past B (infeasible); for $\lambda \geq 1.5$ they sum to value at most $2 < 3$. No scalar λ implements the optimum; Algorithm 1 is exact here precisely because it does not route through a price. What transfers to every v_i, concave or not, is the reformulation and Corollary 5's guarantee — not a market interpretation of the multiplier.

The decomposition also yields the one positive guarantee we have. It says that the difficulty of the multi-receiver problem is concentrated entirely in the single-receiver subproblems: whatever can be achieved for one receiver transfers, without loss, to n of them.

Corollary 5. *This is a computational oracle in the algorithmic sense, unrelated to the gold-support policy Table 3 calls "oracle"; suppose for each receiver i we have one that, given a level b, returns a set $\widetilde{S}_i(b)$ with $c(\widetilde{S}_i(b)) \leq b$ and $u_i(\widetilde{S}_i(b)) \geq (1-\varepsilon)\, v_i(b)$. Then Algorithm 1, run on the curves $\widetilde{v}_i(b) = u_i(\widetilde{S}_i(b))$, returns a feasible allocation of value at least* $(1-\varepsilon)$ OPT.

Proof. Let $(b_i^\star)$ maximise $\sum_i v_i(b_i)$ subject to $\sum_i b_i \leq B$; by Proposition 4 its value is OPT. The algorithm returns $\max\{\sum_i \widetilde{v}_i(b_i) : \sum_i b_i \leq B\}$, which is at least $\sum_i \widetilde{v}_i(b_i^\star) \geq (1-\varepsilon) \sum_i v_i(b_i^\star) = (1-\varepsilon)$ OPT. Feasibility and attainment hold because each $\widetilde{v}_i(b)$ is realised by an actual set of cost at most b, and $\widetilde{v}_i(b) \geq 0$ since $\emptyset$ is always available. □

Remark 4. The technique is standard resource allocation, credited as such; what is worth recording is the conclusion, under two standing assumptions. Costs are integer (Algorithm 1's table is indexed by $b = 0, \ldots, B$), so running time is pseudo-polynomial in B, not polynomial in its bit length. And the oracle hypothesis must hold at *every* level b, not just the optimum: with $\rho_i \equiv 0$ and rel_i additionally modular, each v_i is an ordinary knapsack value function with a $(1-\varepsilon)$-oracle for every $\varepsilon > 0$; for general monotone submodular rel_i, only a $(1-1/e)$-oracle exists, and the Corollary transfers that coarser ratio instead. Either way the n-receiver problem is no harder to approximate than the 1-receiver one, so alongside Proposition 1: the seed-set formulation is off by a factor n or worse, while the correct formulation costs nothing extra in approximability. The reason to abandon the one-set object is not tractability.

Example 2. In the running example, v_1, v_2 jump from 0 to 1 at $b = 1$; v_3 stays at 0 until $b = 2$, then jumps to 1. At $B = 4$ the DP splits $(1,1,2)$ for value 3; a uniform split ($b_i = 4/3$, rounding to 1 each) scores only 2, wasting a document on agent 3 that it cannot use alone. A greedy density rule also scores 2: at the moment it must choose, agent 3's first document has marginal value 0 and loses to agent 1's or agent 2's marginal value 1, and a zero-marginal item, once passed over, is never revisited. The DP avoids this because v_3 is built in full before any budget is committed.

Algorithm 1 Marginal allocation, given any per-receiver oracle

Require: receivers $1..n$, pool F, integer costs, budget B, and an oracle returning $(V_i(b), S_i(b))$ for every i, b — exact ($V_i{=}v_i$, Proposition 4) or $(1-\varepsilon)$-approximate ($V_i{=}\widetilde{v}_i$, Corollary 5)

for $i = 1$ to n **do**
  for $b = 0$ to B **do**
    $(V_i(b), S_i(b)) \leftarrow \mathrm{oracle}(i, b)$
  end for
end for
$A_0(b) \leftarrow 0$ for all b
for $i = 1$ to n **do**
  $A_i(b) \leftarrow \max_{0 \leq t \leq b} A_{i-1}(b-t) + V_i(t)$ ▷ for all $b \leq B$
end for
recover the levels (b_i) attaining $A_n(B)$ **return** $(S_1(b_1), \ldots, S_n(b_n))$

Concavifying v_i (time-sharing between two integer levels) does restore a market-clearing price, at the cost of a randomized allocation; whether a deployed system would accept that trade is a question about the system, not about the theorem, and we do not take a position on it here.

The second loop is $O(nB^2)$ arithmetic operations, or $O(nB)$ with the usual running-maximum bookkeeping when the v_i are concave; either way it is pseudo-polynomial in the budget and linear in the number of receivers, so the coupling across receivers is not where the cost lies. The first loop is the hard part, by Proposition 2. This is the sense in which the per-receiver object is well structured: the multi-receiver coupling is a scalar and the difficulty is local.

Two algorithm families follow, and we are explicit about what is new.

An exact dominance test. No optimal allocation contains an item $f \in S_i$ with

$$\Delta\mathrm{rel}_i(f \mid S_i \setminus \{f\}) \;<\; \rho_i\big(\mathrm{conf}_i(S_i)\big) - \rho_i\big(\mathrm{conf}_i(S_i \setminus \{f\})\big). \quad (4)$$

This is exact, not a first-order approximation: if (4) holds, deleting f raises u_i, because the relevance lost is smaller than the saturation charge *exactly* avoided, and simultaneously frees $c(f) > 0$ of budget, which can only weaken the constraint. The resulting allocation is feasible and strictly better, contradicting optimality. Unlike a cardinality cap, the test restricts the search space without restricting the objective, and it needs no assumption on ρ_i's shape — not even continuity.

A density-greedy heuristic. Building a set that satisfies (4) still requires deciding an admission *order*. The standard device, which we credit rather than claim (distorted-greedy [12], Algorithm 1 of [22], applied to a single agent with a bicriteria guarantee by [4]), ranks candidates by relevance per token, $\Delta\mathrm{rel}_i(f \mid S)/c(f)$, and admits while a threshold is cleared. Turning (4) into a per-token threshold requires relating the exact penalty step to $c(f)$, and this is where the model's own distinction between billed cost and confusable load (§3) has to be specialized to say anything operational: writing $\mathrm{conf}_i(S) := c(S)$ — confusable load approximated by token load, for lack of a confusability oracle — turns (4) into a same-unit condition on $c(S_i)$, and a first-order approximation of *that* gives the per-token rule $\Delta\mathrm{rel}_i(f \mid S)/c(f) \geq \rho_i'(c(S))$ our density-rule baseline (§5) implements. Two approximations are stacked here, not one: the token-load proxy for confusability, and the first-order step, which additionally needs ρ_i differentiable where the measured curves are S-shaped. We report the baseline's results as a heuristic under both approximations, not as an instantiation of the exact test. A per-receiver threshold, exact or approximate, also has no view of what a token would buy at another receiver — which is exactly what Algorithm 1 adds.

Marginal allocation. The objective is separable across receivers and coupled only by the shared budget, so the problem factors: build each receiver's value-versus-budget curve $v_i(b)$ and solve $\max \sum_i v_i(b_i)$ subject to $\sum_i b_i \leq B$ by dynamic programming. This equalizes marginal value per token *across* receivers, which neither a uniform per-receiver rule nor a per-receiver density rule can do — the former spends identically on every receiver, the latter has no view of what a token would buy elsewhere. The saturation penalty enters only through the chosen level, so it may take any shape, including the non-convex ones the measurements suggest. §5 reports that this does not, in fact, outperform a tuned uniform rule on our benchmark, and why.

5 Experiments

Setup. We use MuSiQue-Ans [24], where the structure we need is annotated: each instance supplies 20 candidate paragraphs with support labels, and sub-questions each carrying the index of the paragraph answering it. Receivers are sub-questions, the item pool is the paragraph set, and cost is tokens transmitted summed over receivers. Ground-truth relevance being labeled makes an oracle allocation computable. As §4.3 flags, we take $\mathrm{conf}_i(S) := c(S)$ throughout, i.e. token load stands in for confusable load; utility is proxied by exact-match accuracy, since rel_i and ρ_i are not separately observable outcomes. Neither substitution is assumed elsewhere: every theory result in §4.1–§4.3 holds for general conf_i and u_i.

Splitting MuSiQue's single-agent task across per-sub-question receivers is our framing (§6.1 discusses this and contamination as threats to validity); we restrict to the branching families where two sub-questions carry no back-reference and are genuinely independent. Comparisons are paired — every policy answers the same items — using McNemar's exact test on discordant pairs and, for the token–accuracy points, a bootstrap over items.

Protocol. Two instruction-tuned models of the same family were used, at temperature 0, with a fixed answer-extraction template shared by every arm so that formatting cannot differ between policies. The matched-budget grid has 12 arms over 60 instances (1440 calls, all completed); the earlier 4-arm grid covers 69 instances at $n = 138$ receiver-instances. Every arm sees an identical prompt scaffold and differs only in which paragraphs the receiver is given, and arms are matched on delivered tokens before accuracy is compared, since an accuracy gain bought with a larger budget is not a gain from the mechanism. Runs are keyed and resumable per (instance, policy, arm, receiver), and duplicate keys are removed before analysis; without this the pairing silently breaks and n inflates.

Table 2: Paired comparisons, $n = 120$ receiver-instances (60 instances $\times\, 2$ independent sub-questions), matched-budget grid. Δ is the first policy minus the second, with a 95% paired bootstrap CI; token ratio > 1 means the first is cheaper. Starred $p < 0.05$ (McNemar, exact); see §6.1 for a cluster-robust check.

Comparison	Δacc	95% CI	p	tok.
per-recv. vs. seed set	+0.192	$[+.10, +.28]$	0.0004*	1.1×
oracle vs. broadcast	+0.050	$[-.01, +.11]$	0.146	21.4×
density (loose) vs. b'cast	−0.042	$[-.10, +.01]$	0.227	2.7×
uniform k=5 vs. b'cast	−0.083	$[-.14, -.03]$	0.013*	5.4×
density: tight vs. unif. k=5	+0.000	$[-.04, +.04]$	1.000	0.8×

Table 3: Accuracy and cost per policy, matched-budget grid, $n = 120$. Tokens are per receiver; the ratio is broadcast's mean tokens divided by the policy's, from the two numbers shown. Absolute exact-match is suppressed for deltas since MuSiQue predates current models (§6.1: 0% memorised here, up to 5% on the other model). "Density" is §4.3's exact-dominance heuristic at two thresholds, not Algorithm 1, evaluated separately in Table 5.

Policy	tokens	ratio	Δacc
broadcast (all items)	2256	1.0×	—
oracle (labelled)	106	21.3×	+0.050
per-recv., unif. k=5	420	5.4×	−0.083
density, tight q=0.7	511	4.4×	−0.083
density, loose q=0.3	846	2.7×	−0.042
seed set, k=3	251	9.0×	−0.392
random, k=3	336	6.7×	−0.667
no context	0	—	−0.792

The separation appears in the data. The strongest measured effect is the first row of Table 2: per-receiver allocation beats a seed-set baseline — items ranked by aggregate score and sent to everyone — by 19.2 accuracy points at matched cost. Proposition 1 predicts the gap should grow with the number of receivers, and it does: as we fuse instances to raise n from 2 to 8 to 16, seed-set delivery recall falls from 0.45 to 0.08 to 0.01 while per-receiver recall stays above 0.55.

A cost–quality frontier, honestly reported. The loose density rule cuts tokens 2.7× at an observed −4.2 accuracy points, 95% CI $[-0.10, +0.01]$ (row 3): the point estimate favours broadcast, and the sample cannot rule out a real loss in this range, but it also cannot distinguish −4.2 points from zero. We report the estimate and interval rather than a verdict; we did not run a pre-registered non-inferiority test, so "indistinguishable from free" is the honest description, not "free." Pushing to 5.4× does cost 8.3 points, and

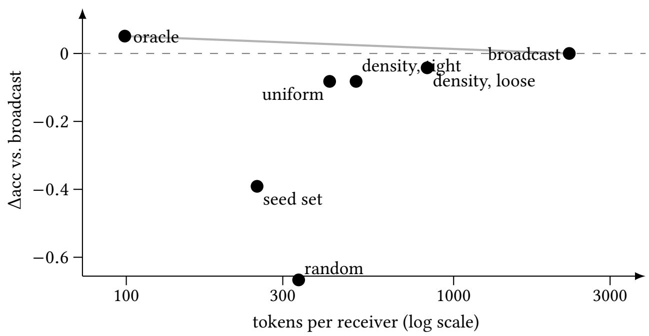

Figure 2: The measured cost–quality frontier, $n = 120$ receiver-instances. The dashed line is broadcast's accuracy; anything above it is both cheaper *and* better. The oracle sits there, at 21× fewer tokens, which is the order of magnitude the problem makes available. The implementable policies cluster well below the grey envelope, and — the point of the paper's negative result — they differ from one another by far less than any of them differs from the oracle. The seed set at matched cost is 39 points down, and random delivery (−0.667) fixes the scale of the task. An empty-allocation arm (not shown, 0 tokens, −0.792) is the contamination control.

Table 4: Proposition 1 predicts the seed-set gap grows with the number of receivers. Delivery recall as instances are fused to raise n. The seed set collapses; per-receiver allocation does not.

receivers n	2	8	16
seed set (matched cost)	0.45	0.08	0.01
per-receiver, uniform	0.67	0.56	0.56

the CI here excludes zero (row 4), so the frontier is real. The oracle beats broadcast by an observed ∼ 5 points at 21× fewer tokens, consistently in sign across two models (+0.051 and +0.050) but with a CI excluding zero on only one of them — a consistent effect at borderline power, not an established one.

A negative result on algorithm design. Across four value-curve designs, marginal allocation does not beat a tuned uniform per-receiver rule at matched budget; nor does the density rule (row 5, exactly zero at $p = 1.000$ while spending 25% more). We also found that replacing lexical relevance with a dense embedding scorer makes per-receiver retrieval *worse* (0.43 → 0.25 top-1 recall), because the sub-questions are terse relation templates whose entity identity a dense encoder discards — while the same change nearly doubles seed-set recall, which needs only instance-level topicality.

Composite receivers manufacture the heterogeneity Proposition 1 draws its advantage from: the full multi-hop question needs *all* its supporting paragraphs where a sub-question needs one, so a global k must starve one receiver or overfeed the other. Even then, marginal allocation only matches uniform, because it can only help where per-receiver value curves are *distinguishable from label-free signals*, and here they are not: every receiver draws from a common

Table 5: Four designs for the value curve $v_i(b)$ that Algorithm 1 consumes, scored by delivery recall at matched spend against a tuned uniform per-receiver rule, $n = 130$ composite instances (SE ≈ 0.04 on a recall near 0.6, comparable to the differences below — read "ties" as *not detectably different*, not *proven equal*). The structure Proposition 4 exposes is real; on this benchmark it is not *estimable.*

Value curve $v_i(b)$	Outcome vs. tuned uniform
sum of relevance scores	**loses**: near-linear in b, nothing to balance
noisy-OR over scores	ties; +4–5pp at large budgets only
captured own score mass	ties
any of the above, composite receivers	ties (≈ 11% cheaper at equal recall)

Table 6: Replacing lexical relevance with a dense embedding scorer *harms* per-receiver retrieval but helps the seed set. The two signals fail in opposite directions: embeddings capture what an instance is about, lexical matching captures which receiver needs what.

Delivery recall	lexical	embedding
per-receiver, top-1	**0.43**	0.25
per-receiver, top-5	**0.79**	0.61
seed set, k=3	0.45	**0.85**

pool with a similar score distribution, so an equal split is already near optimal. Heterogeneity in what receivers *need* did not surface as heterogeneity in what could be *estimated* about them.

The direction of that failure is legible in the data rather than mysterious. MuSiQue's branching sub-questions are mostly terse relation templates — of the form `"<entity> » <relation>"`, for instance `"The Poor Boob » screenwriter"` — and not natural-language questions. Exact entity-token overlap locates the right paragraph; a dense encoder maps such a string into a topical neighbourhood and discards the entity identity that distinguishes one receiver's need from another's. The seed-set row of Table 6 is the control that confirms this: it moves the other way, 0.45 → 0.85, because sending one global set to everyone requires only instance-level topicality, which is exactly what a dense encoder captures well. So the two signals fail in opposite directions — embeddings know what an instance is about but not who needs what; lexical matching discriminates between receivers but misses paraphrase. The residual gap is therefore attributable to the *query representation* rather than to the scorer family, and a hybrid or entity-aware scorer is the obvious next attempt. We have not tried it, and we would not expect it to close the gap entirely.

Figure 2 puts all of this on one axis. The visual summary is the paper's argument in miniature: the gap between the implementable policies and the oracle dwarfs the gaps among the policies themselves, which is what it looks like when the binding constraint is not the allocation rule.

Three independent observations thus point the same way: the binding constraint is relevance *estimation*, not allocation *optimization*. On the composite-receiver design just described, the oracle fixes the scale of it, reaching full delivery recall on 208 tokens where the best implementable policy reaches only 62% recall on 929 — a different sample and metric from Table 3's accuracy grid, and not directly comparable to its 106-versus-broadcast tokens. No allocation algorithm closes this gap, because the information required is not present in the scores.

Logged traces. Two individual calls from the matched-budget grid, tracing an over-delivery failure and a seed-set failure item by item down to the model's own answer, are in the supplementary material rather than here, to leave room for the analyses above.

6 Discussion and conclusion

6.1 Discussion

The results divide cleanly. The formulation-level claims are strong: the seed-set object is wrong by a factor n even on monotone submodular instances, unboundedly wrong once saturation is added, and both the classical toolchain and its nearest general successor are unavailable here. The algorithm-level claims are deliberately weak: we do not have a method that beats a tuned uniform per-receiver rule, and we report that rather than selecting a favorable budget point.

We take the negative result to be informative rather than merely disappointing (§5's diagnosis: heterogeneity in what receivers *need* did not surface as heterogeneity in what could be *estimated* about them here). Settings where it does — receivers with genuinely different and predictable appetites — are where the separation's practical value should be sought.

Threats to validity. We list these in the order we think a referee should weigh them.

The multi-receiver framing is ours. Splitting MuSiQue into per-sub-question receivers is an interpretation, not a property of the dataset — defensible, since the sub-questions and supporting paragraphs are annotated, which is what makes the oracle computable, but no conclusion here is a measurement on a deployed multi-agent system. We restrict to branching decompositions because sequential ones would make receivers dependent and the framing dishonest.

Contamination. The benchmark predates current models: 0% of receiver-instances in Table 3's run are answered correctly with no context, rising to 5% on the n=138 grid of §5's Protocol paragraph. We report policy deltas rather than absolute exact-match throughout; this prevents memorisation from being mistaken for policy quality, provided it affects arms equally.

Statistical power. The oracle-over-broadcast effect is +0.051 on one model, +0.050 on the other, significant on only one (p = 0.039 vs. 0.146) — a consistent ∼ 5-point effect at borderline power, not an established one.

Clustering. Each original instance contributes two receivers sharing a pool and difficulty, so treating all 120 as independent risks understating variance. A block bootstrap over the 60 instances and a cluster permutation test leave three of five rows unchanged to three decimals, widening the other two only slightly (uniform k=5 vs. broadcast: $p = 0.013 \to 0.028$), crossing 0.05 nowhere. We report the flat analysis throughout; larger or more heterogeneous clusters would need the block version by default.

Calibration of ρ. The saturation penalty's shape comes from published degradation curves whose magnitude is strongly model- and task-dependent, and at least one study finds large models essentially unaffected by non-confusable filler. Proposition 1(a) needs no degradation at all; the empirical claim about over-delivery is only as general as that literature. Neither proposition supplies an approximation ratio for non-trivial ρ_i, and Corollary 5 supplies one only conditional on a single-receiver oracle — the unconditional ratio is Open Problem (i) below.

Open problems. (i) An unconditional approximation guarantee for Definition 1 with non-trivial ρ_i, which Corollary 5 reduces to the single-receiver case where u_i is neither monotone nor submodular; whether saturation adds hardness beyond Proposition 2's knapsack structure is open too. (ii) Bundle identification: our complementarity witness assumes the jointly-decisive pair is known, whereas a system would have to discover it. (iii) An end-to-end multi-agent task on a dependent communication graph, evaluated against stronger receiver-aware routing baselines than the global-top-k translation of the seed-set idea used here. (iv) Endogeneity: we treat the receiver set as given, but a deployed orchestrator decides which agents to instantiate in response to the task, and we see no route to a competitive-analysis treatment that does not first fix a model of how it responds.

6.2 Conclusion

Much inter-agent context traffic looks removable at no accuracy cost we could measure, on a controlled proxy under paired testing — not, without further work, on a deployed multi-agent system. A 2.7× reduction has an observed cost indistinguishable from zero at this sample size, and an oracle reaches 21× with an observed accuracy gain, so the ceiling is plausibly an order of magnitude rather than a few percent.

Reaching that ceiling, the obstruction is not where we expected: not degradation under irrelevant context, nor item interaction, though both occur; it is that the seed-set object — one set serving a network, the natural reading of influence maximization — is off by a factor n from the per-receiver object even on well-behaved instances, unboundedly once saturation is added. The nearest two toolchains do not apply, the second because its parameters do not exist rather than its bound being loose. The switch of object costs nothing in approximability under an explicit oracle assumption, as a dynamic program though, not in general a decentralizable price.

The remaining order of magnitude runs through better estimates of what a receiver needs, not better allocation algorithms. Practically: switch to per-receiver allocation, worth 19 points over the globally-scored alternative, then spend further effort on retrieval.

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